Complete Kubernetes Observability Architecture (Logging, Metrics, Alerting)

1. Introduction

This document provides a complete production-ready observability setup for Kubernetes microservices using Fastify, ELK stack (Elasticsearch, Logstash/Filebeat, Kibana), Prometheus, Grafana, and Alertmanager. It explains logging, metrics collection, monitoring, and alerting, along with Helm-based deployment.

2. Architecture Overview

There are three major pillars:
1. Logging (ELK Stack)
2. Metrics (Prometheus + Grafana)
3. Alerting (Alertmanager + Slack)

3. Logging Architecture

Fastify uses Pino logger to produce structured JSON logs. These logs are written to stdout in containers. Filebeat (DaemonSet) collects logs from all pods and sends them to Elasticsearch. Kibana is used for search and visualization.

Logging Flow

Fastify → stdout → Filebeat → Elasticsearch → Kibana

4. Metrics Architecture

Metrics represent numerical measurements such as CPU, memory usage, request latency, and error rates. Node Exporter collects node metrics, cAdvisor collects container metrics, and application metrics are exposed via /metrics endpoint using prom-client. Prometheus scrapes and stores these metrics.

Metrics Flow

Node Exporter + cAdvisor + Fastify (/metrics) → Prometheus → Grafana

5. Monitoring Application Metrics

To monitor application metrics in Fastify, use prom-client to expose metrics such as request count, response time, and errors.

Example Fastify Metrics Code

```
const client = require('prom-client');
const register = new client.Registry();
client.collectDefaultMetrics({ register });

fastify.get('/metrics', async (req, reply) => {
  reply.header('Content-Type', register.contentType);
  return register.metrics();
});
```

6. Alerting Architecture

Prometheus evaluates alert rules. When conditions are met, alerts are sent to Alertmanager, which forwards notifications to Slack.

Alert Flow

Prometheus → Alertmanager → Slack

7. Helm Deployment

Helm simplifies deployment in Kubernetes.

Install ELK Stack

```
helm repo add elastic https://helm.elastic.co
helm install elasticsearch elastic/elasticsearch
helm install kibana elastic/kibana
helm install filebeat elastic/filebeat
```

Install Monitoring Stack

```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm repo update
helm install monitoring prometheus-community/kube-prometheus-stack
```

8. Difference Between Logs and Metrics

Logs are detailed event records (text/json) that describe what happened at a specific point in time. Metrics are numerical values aggregated over time used to monitor system health.

Logs

- Event-based
- High detail
- Used for debugging
- Example: error message, request log

Metrics

- Numeric time-series
- Aggregated
- Used for monitoring and alerting
- Example: CPU usage, request rate

9. Best Practices

- Use structured logging (JSON)
- Expose metrics endpoint
- Use Helm charts for deployment
- Enable persistent storage
- Secure services with TLS and RBAC

10. Conclusion

This architecture provides full observability: logs, metrics, and alerting. It enables debugging, performance monitoring, and proactive incident response.